LEFA LAZY MALATJI



SOFTWARE DEVELOPER | AI & NLP RESEARCHER

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PROFESSIONAL SUMMARY

Software developer with one year of internship experience building web applications and managing PostgreSQL databases. Computer Science Honours graduate, MSc candidate and published NLP researcher focused on low-resource Sepedi-English text generation. Combines C#/ASP.NET Core and PHP development with relational database design, statistics and practical problem solving.

PROFESSIONAL EXPERIENCE

Software Developer Intern | Simplifon

One year of professional software development experience

- Developed and maintained web-based applications.

- Developed and managed PostgreSQL databases supporting web applications.

Lecturer | Rosebank College

- Teach programming, software development and database concepts through practical examples and assessments.

- Support learning in C#, object-oriented programming, web development, SQL and database normalisation.

TECHNICAL SKILLS

Software: C#, ASP.NET Core Web API, MVC, PHP, JavaScript, HTML5, CSS3, Bootstrap 5.
Databases: PostgreSQL, SQL Server, MySQL, SQL, data modelling, relational design, normalisation and Entity Framework migrations.
AI & data: Python, TensorFlow/Keras, NLP, text preprocessing, FNet, LSTM, mBERT research, BLEU/ROUGE evaluation and statistics.
Tools & practices: Git, GitHub, Docker, REST APIs, Swagger and Agile working environments.

EDUCATION

MSc in Computer Science | University of Limpopo | 2026-present

In progress. Investigating data-scarcity effects on Sepedi-English text generation using FNet, LSTM and mBERT.

BSc (Honours) in Computer Science | University of Limpopo | 2025

Advanced Operating Systems, Computer Security, Advanced Databases, Speech Technology and Research Project.

BSc in Mathematical Sciences (ECP) | University of Limpopo | 2021-2024

Completed. Majors: Computer Science and Statistics.

PROJECTS & RESEARCH

Selected work in intelligent applications, software and data systems

Store Hub | E-Commerce Platform

PHP | PostgreSQL | Multi-role application | AI features

- Built a multi-role e-commerce platform with AI-driven personalisation and intelligent product search.

- Integrated AI business intelligence reporting and a session-aware customer-support chatbot.

GitHub: Advanced E-Commerce Marketplace Platform

HelpDeskHub | In Development

C# | ASP.NET Core REST API | PostgreSQL | Entity Framework | Identity

- Developed ticket, department and user API functionality, using DTOs and Swagger for API development and testing.

- Integrated ASP.NET Identity and database migrations, linking tickets to creators, assigned users, comments and history.

- Roadmap includes an MVC front end, workload-based assignment, notifications and reporting.

Pre-School Management System

Admin dashboard | Data management | CRUD

- Built admin features with dashboard statistics, secure login and management of learners, teachers, books and borrowing records.

GitHub: Pre-School Management System

Airport Sky Base System

Application development | Information management | Testing

- Developed and tested an airport management application with an interface for accessing and managing airport information.

GitHub: Airport Sky Base System

Burzarsnet | Team Project

Web platform | Admin management | Team of four

- Collaborated with three team members to develop a South African bursary and internship web platform with admin management.

Website: burzarsnet.com

PUBLISHED RESEARCH

Lefa L. Malatji & Thipe I. Modipa. Investigating Data Scarcity and Model Adaptation for Sepedi-English Code-Switching Text-Generation.

Research on generative AI for code-switching using the FNet architecture. Current MSc work extends the investigation to data-scarcity effects and model comparison.

ADDITIONAL INFORMATION

Languages: Sepedi and English. References available on request.